Fiscal Year:	FY 2023	**Task Last Updated:**	FY 07/13/2023
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Project Title:	The Effect of Isochoric Supercooling on Human Liver Metabolic Function		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94143-2205	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 BRASH2101: Translational Research Institute for Space Health (TRISH) Biomedical Research Advances for Space Health
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No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
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Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Rubinsky, Boris Ph.D. (University of California, Berkeley)		
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Task Description:	Hibernation and hypothermia have been proposed as potential measures to reduce crew metabolic rate during exploration class space missions, but it remains unclear whether they adequately mitigate the biological challenges posed by extended duration spaceflight. Body temperature is lowered by only a few degrees in hibernating bears and mild therapeutic hypothermia, thus basal metabolic processes and needs persist. In this research, we hypothesize that isochoric (constant volume) supercooling can be used to preserve human tissues at sub-zero temperatures without the damaging effects of freezing. We propose that cell structures will remain aqueous in sub-zero isochoric supercooled conditions, while all cell functions, constructive or destructive, will halt. Additionally, due to the lack of ice formation and freeze-thaw, tissues preserved by isochoric supercooling will sustain minimal damage and regain full function upon recovery to normal body temperature. This form of suspended animation precludes ongoing metabolic demands and confers resistance to a host of spaceflight-associated health risks. To test our hypothesis, we propose 2 aims that will first determine the fundamental biology of human tissues preserved by isochoric supercooling and then establish the scalability of the technology to an intact human organ. We chose human liver as the model test tissue because the liver performs a multitude of metabolic functions, serves as a central regulator of whole-body metabolism, and is the largest solid organ in the human body. In Aim 1, we propose to determine the metabolic function of human induced pluripotent stem cell (iPSC)-derived liver organoids during and following recovery from isochoric supercooling. During supercooled preservation, we will assess metabolic activity through functional assays, global metabolomics, and transcriptomics. After recovery, we will determine cell viability, activation of cell death and cell stress pathways, and return to metabolic function. In Aim 2, we propose to test the scalability of isochoric supercooling to preserve an intact human liver, using organs deemed not suitable for transplantation obtained through the transplant donor network. We will compare the metabolic function, histology, metabolomic profile, and global gene expression of whole intact livers before and after isochoric supercooled preservation for various durations of time. If successful, the results of this research will demonstrate isochoric supercooling as a potential cross-cutting countermeasure that can significantly reduce crew metabolic demands and mitigate multiple spaceflight-associated health risks.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The impact of our research is that our findings may in the near-term improve storage and transport of donor organs for transplantation, and in the long-term mitigate risks for human health during prolonged spaceflight.
Task Progress:	Hibernation and hypothermia have been proposed as potential measures to reduce crew metabolic rate during exploration class space missions, but it remains unclear whether they adequately mitigate the biological challenges posed by extended duration spaceflight. Body temperature is lowered by only a few degrees in hibernating bears and mild therapeutic hypothermia, thus basal metabolic processes and needs persist. In this research, we hypothesize that isochoric (constant volume) supercooling can be used to preserve human tissues at sub-zero temperatures without the damaging effects of freezing. We propose that cell structures will remain aqueous in sub-zero isochoric supercooled conditions, while all cell functions, constructive or destructive, will halt. Additionally, due to the lack of ice formation and freeze-thaw, tissues preserved by isochoric supercooling will sustain minimal damage and regain full function upon recovery to normal body temperature. This form of suspended animation precludes ongoing metabolic demands and confers resistance to a host of spaceflight-associated health risks. To test our hypothesis, we propose 2 aims that will first determine the fundamental biology of human tissues preserved by isochoric supercooling and then establish the scalability of the technology to an intact human organ. We chose human liver as the model test tissue because the liver performs a multitude of metabolic functions, serves as a central regulator of whole-body metabolism, and is the largest solid organ in the human body. In Aim 1, we propose to determine the metabolic function of human induced pluripotent stem cell (iPSC)-derived liver organoids during and following recovery from isochoric supercooling. During supercooled preservation, we will assess metabolic activity through functional assays, global metabolomics, and transcriptomics. After recovery, we will determine cell viability, activation of cell death and cell stress pathways, and return to metabolic function. In Aim 2, we propose to test the scalability of isochoric supercooling to preserve an intact human liver, using organs deemed not suitable for transplantation obtained through the transplant donor network. We will compare the metabolic function, histology, metabolomic profile, and global gene expression of whole intact livers before and after isochoric supercooled preservation for various durations of time. If successful, the results of this research will demonstrate isochoric supercooling as a potential cross-cutting countermeasure that can significantly reduce crew metabolic demands and mitigate multiple spaceflight-associated health risks. During the first year of the project, we have shown the hepatic cell line (HepG2) organoids are preserved with high viability in isochoric supercooling after 48 hours. In addition, pig livers are preserved with intact tissue architecture in isochoric supercooling after 48 hours. In the coming year, we will focus on 1) optimization of liver organoid and whole liver isochoric chambers, 2) viability of HepG2 organoids through timecourse, 3) metabolic function of iPSC-derived liver organoid in isochoric supercooling, and 4) demonstration of perfusion and isochoric perfusion of whole human liver.
Bibliography Type:	Description: (Last Updated: 11/17/2025)